# Sonu Kumar

Bachelor of Technology
Electrical Engineering
National Institute of Technology, Patna

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## Education

| Degree/Certificate | Institute/Board | CGPA/Percentage | Year |
|---|---|---|---|
| B.Tech., EE | National Institute of Technology, Patna | 7.82 | 2023-2027 |
| Senior Secondary | BSEB | 91.2 | 2022 |
| Secondary | CBSE | 88.6 | 2020 |

## Experience

- **Research Project – Transmission Line Inspection Robot** *NIT Patna*
  *Power Systems, Robotics & Computer Vision* Dec 2024 – Jun 2025
  - Developed a robotic inspection system for detecting faults in high-voltage transmission lines.
  - Applied vision-based techniques to support predictive maintenance in power transmission networks and improve inspection safety.
- **5G-Enabled Disaster Response Drone – 5G Innovation Hackathon 2025**
  *DoT Sponsored National Hackathon (Top 25–50 Teams)* March 2025 – Sep 2025
  - Developed a **5G-enabled aerial system** for real-time monitoring and low-latency data transmission.
  - Integrated sensors and telemetry enabling reliable field-level data acquisition.

## Projects

- **ANAV Autonomous Drone – ISRO IRoC-U 2025**
  *Embedded Systems, Sensors, Control*
  - Led a team in IRoC-U Challenge 2025, developing a functional drone that advanced through two rounds.
  - Designed an autonomous drone system integrating sensors and control logic for stable flight and infrastructure inspection.
  - Worked on sensor integration and real-time data acquisition for aerial inspection.
- **Hand Gesture Controlled Robot**
  *OpenCV, MediaPipe, ESP32*
  - Built a real-time **gesture-controlled robotic platform** using computer vision and embedded control.
  - Implemented **wireless motion control** via ESP32 for responsive and reliable robot operation.

## Skills

- **Programming**: C, C++, Python, MATLAB
- **Core Electrical**: Power Systems, Control Systems, Electrical Machines
- **Hardware IoT**: Arduino (Uno, Mega), ESP32, NodeMCU8266, Raspberry Pi, NVIDIA Jetson Nano
- **Soft Skills**: Leadership, Teamwork, Communication, Project Management

## Certifications

- **IIT Patna**, DroneX – The Drone Challenge *Jun 2024*
- **Skill India**, Kisan Drone Operator *May 2024*
- **NIT Patna**, Laser Bot Race Challenge *Jan 2024*

## Achievements

- Ranked 3rd in district, Intermediate Science (2022)
- Won Gold in Laser Bot Race, Silver in Tinkering Lab, and Bronze in Grand Line Challenge

## Positions of Responsibility

- **Vice President,** Robotics Club, NIT Patna *2025 – Present*
- **Technical Team Member,** Robotics Club, NIT Patna *2023 – 2025*